Yusei Ito

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Experiences

January 2026 – April 2026: **Visiting research student, MBZUAI**, Abu Dhabi, UAE
Supervisor: Prof. Martin Takáč
Published a paper in ICML2026 workshop [4].

August 2025 – November 2025: **Research Internship, Toyota Motor Corporation**, Tokyo
Supervisor: Dr. Yuta Suzuki & Dr. Masaki Adachi
Submitted a journal paper and it is currently under review.

May 2024 – September 2024: **Research Internship, OMRON SINIC X Corporation**, Tokyo
Supervisor: Dr. Tatsunori Taniai & Dr. Ryo Igarashi
Published a paper in ICLR2025 [6].

April 2023 – March 2025: **Project Researcher (part time), Osaka University**, Osaka

Education

Osaka University, Japan (April 2019 – present)

April 2025 – present: **Ph.D. course of Engineering** in Applied Physics
Supervisor: Prof. Kanta Ono

March 2025: **Master of Engineering** in Applied Physics
Supervisor: Prof. Kanta Ono

March 2023: **Bachelor of Engineering** in Applied Physics
Supervisor: Prof. Kanta Ono

Publications

◆ **Journals**

[1] Yusei Ito, Naoya Chiba*, Tatsunori Taniai*, Ryo Igarashi, Yuta Suzuki, Kotaro Saito, Yoshitaka Ushiku, Kanta Ono, "Enabling Single-Observation Decomposition of Multi-phase X-ray Diffraction Patterns via Generative Deep Learning", npj Computational Materials **12**, 238 (2026).

[2] Yusei Ito, Yasuo Takeichi, Hideitsu Hino, Kanta Ono, "Optimal spectroscopic measurement design: Bayesian framework for rational data acquisition", Machine Learning: Science and Technology **6**, 025037 (2025).

[3] <u>Yusei Ito</u>, Yasuo Takeichi, Hideitsu Hino, Kanta Ono, "Rational partitioning of spectral feature space for effective clustering of massive spectral image data", Scientific Reports **14**, 22549 (2024).

◆ **International Conference (Refereed)**

[4] <u>Yusei Ito</u>, Aidar Alimbayev*, Klea Ziu*, Deepak Kumar*, Kanta Ono, Martin Takáč, "Revisiting the Form of Attention with Positional Encoding for Molecular Structures", ICML 2026 Workshop on AI for Physics, Seoul, South Korea, July (2026). (Acceptance rate: 99/194 = 51.0%)

[5] <u>Yusei Ito</u>, Yuta Suzuki, Tomoya Murata, Masaki Adachi, "LapidaryEngine: Feedback-Guided Iterative Text-to-Crystal Structure Generation", AI4X - Accelerate Conference 2026, Singapore, June (2026).

[6] <u>Yusei Ito*</u>, Tatsunori Taniai*, Ryo Igarashi, Yoshitaka Ushiku, Kanta Ono, "Rethinking the role of frames for SE(3)-invariant crystal structure modeling", The Thirteenth International Conference on Learning Representations (ICLR 2025), Singapore, April (2025). (Acceptance rate: 32.1%)

[7] Yuki Nishihori, <u>Yusei Ito</u>, Yuta Suzuki, Ryo Igarashi, Yoshitaka Ushiku, Kanta Ono, "Transformer as a Neural Knowledge Graph" , AI for Accelerated Materials Design-ICLR2025 Workshop, Singapore, April (2025).

[8] <u>Yusei Ito</u>, Yasuo Takeichi, Hideitsu Hino, Kanta Ono, "Optimal Spectroscopic Measurement Design: Bayesian Framework for Rational Data Acquisition", AI for Accelerated Materials Design-NeurIPS2024 Workshop, Vancouver, Canada, December (2024).

◆ **International Conference (Unrefereed)**

[9] Yasuo Takeichi, Shuta Nakao, <u>Yusei Ito</u>, Yasuhiro Niwa, Masao Kimura, Kanta Ono, "Mineral phase distribution and reduction degradation behavior of iron ore sinters investigated by gigapixel imaging XAFS", The 2nd International Symposium on Iron Ore Agglomerates (SynOre 2026), Toyama, Japan, October (2026).

[10] Yasuo Takeichi, Shuta Nakao, <u>Yusei Ito</u>, Yasuhiro Niwa, Masao Kimura, Kanta Ono, "Mineral phase identification in iron ore sinters exploiting intrinsic polarization dependence of X-ray absorption spectromicroscopy", The 19th International Conference on X-ray Absorption Fine Structure (XAFS2026), Chiang Mai, Thailand, June (2026).

[11] Kai Kawasaki, Yusaku Nakajima, <u>Yusei Ito</u>, Yasuo Takeichi, Kanta Ono, "Single-Particle Chemical State and Kinetic Analysis of Mechanochemical Reactions Using Gigapixel X-ray Microscopy", The International Chemical Congress of Pacific Basin Societies 2025 (Pacifichem 2025), Honolulu, Hawaii, December (2025).

[12] <u>Yusei Ito</u>, Yasuo Takeichi, Hideitsu Hino, Kanta Ono, "Gigapixel X-ray Spectromicroscopy Data Analysis by Clustering", 16th International Conference on X-Ray Microscopy (XRM 2024), Lund, Sweden, August (2024). (oral presentation)

[13] <u>Yusei Ito</u>, Yasuo Takeichi, Hideitsu Hino, Kanta Ono, "Optimal Spectromicroscopic Experimental Design for Massive Data Acquisition", 16th International Conference on X-Ray

Microscopy (XRM 2024), Lund, Sweden, August (2024).
[14] Yasuo Takeichi, Yusei Ito, Yasuhiro Niwa, Masuo Kimura, Kanta Ono, “Development and conventional analysis of gigapixel imaging XAFS”, 16th International Conference on X-Ray Microscopy (XRM 2024), Lund, Sweden, August (2024).
[15] Yusei Ito, Yasuo Takeichi, Hideitsu Hino, Reiko Murao, Kanta, Ono, “Gigapixel X-ray micro-spectroscopy data analysis by using machine learning”, Twenty-Sixth Congress and General Assembly of the International Union of Crystallography (IUCr 2023), Melbourne, Australia, August (2023).
[16] Yasuo Takeichi, Yusei Ito, Yasuhiro Niwa, Reiko Murao, Masao Kimura, Kanta Ono, “Development of gigapixel imaging XAFS”, Twenty-Sixth Congress and General Assembly of the International Union of Crystallography (IUCr 2023), Melbourne, Australia, August (2023).

◆ **Domestic Conference**

23 talks (including 11 as first presenter)

Skills

- **Deep learning algorithm development** in Python and PyTorch (applied in [1][4]–[6])
- **Real world data analysis** using machine learning (applied in [2][3][8])
- **CUDA Programming** for accelerating deep learning algorithm (applied in [6])
- 5+ years of programming experience in **Python**
- Proficient in **Git/GitHub** and **Docker**
- Background in **Physics**, particularly **Materials Science**
- **English** – Business level (TOEIC 815 of 990 in December 2023, Experience in collaborative research projects with international team [4])